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Course: Calculus Beta

Assignment: MyLab Math - Uge 2

Use l'Hôpital's Rule to find the limit.

$$\lim_{x \rightarrow 0^+} \frac{\ln(x^2 + 6x)}{\ln x}$$

Substituting $x = 0$ returns a limit of $\frac{0}{0}$. L'Hôpital's Rule states that if $f(a) = g(a) = 0$, f and g are differentiable on an open interval I containing a , and $g'(x) \neq 0$ on I if $x \neq a$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, assuming that the limit on the right side of this equation exists. Stop differentiating when the limit no longer results in an indeterminate form.

In this case, $f(x) = \ln(x^2 + 6x)$. Find $f'(x)$.

$$f'(x) = \frac{2x + 6}{x^2 + 6x}$$

Now find $g'(x)$ where $g(x) = \ln x$.

$$g'(x) = \frac{1}{x}$$

Divide $f'(x)$ by $g'(x)$ and simplify the resulting expression.

$$\begin{aligned} \frac{f'(x)}{g'(x)} &= \frac{\frac{2x + 6}{x^2 + 6x}}{\frac{1}{x}} \\ &= \frac{2x + 6}{x + 6} \end{aligned}$$

Take the limit by direct substitution.

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{2x + 6}{x + 6} &= \frac{2(0) + 6}{0 + 6} \\ &= 1 \end{aligned}$$

$$\text{Thus, } \lim_{x \rightarrow 0^+} \frac{\ln(x^2 + 6x)}{\ln x} = 1.$$